

9 Chloride short overview

Chloride is required in very small amounts, but avocado is sensitive to accumulation and leaf injury.

Why chloride deserves special attention

- **Essential but rarely deficient.** Plants need chloride for basic cell and photosynthetic functions, yet normal water and inputs usually supply enough.
- **Mobile in soil water.** Chloride is negatively charged and is not strongly held by most soils, so it moves with water.
- **Sensitive crop.** Avocado can accumulate chloride in older leaves, producing tip and margin burn and eventual leaf drop.
- **Source and management both matter.** Irrigation water may supply chloride every cycle, while under-irrigation, poor uniformity, evaporation, or weak drainage can concentrate it.
- **Rootstock matters.** Tolerance and chloride uptake differ among rootstocks, so site history and planting material belong in the diagnosis.

How the main soil and environmental factors affect chloride

Factor	Effect on chloride risk
pH and biology	Usually little direct control of chloride; indirect effects occur through roots structure and water movement.
Organic matter	Affects infiltration evaporation and rooting; compost manure and some products may also add chloride or EC.
Texture	Sandy soil can leach chloride readily but dries quickly; fine soil stores more water but may drain slowly if structure is poor.
Irrigation regime	Long dry intervals concentrate salts; excess frequency reduces air if drainage is weak; nonuniform wetting creates hot spots.
Drainage and leaching	Adequate leaching lowers chloride only when water and salts can move below roots and leave the root zone.
Salinity and sodium	Chloride may contribute to total salinity while sodium can also damage infiltration; measure all three rather than using one as a substitute.

Leaf interpretation

- **Visible pattern.** Older leaves often develop brown tips and margins because they have accumulated chloride over time.
- **Tentative UC guide.** Leaf chloride of about 0.25 to 0.50 percent is listed in the excess range. Treat this as a guide, not a universal injury threshold.
- **Look for look-alikes.** Drought, root rot, sodium toxicity, fertilizer burn, heat, and other stresses can overlap with tip burn.

10 Chloride diagnostic sheet

The goal is to find where chloride enters, where it concentrates, and whether the root zone can safely move it.

Minimum measurements

Measure or observe	Why
Leaf chloride and sodium	Confirms tissue accumulation and helps separate total salt stress from specific ion exposure.
Irrigation water chloride sodium and EC	Defines the continuing source and total salt load; sample each source and blend.
Soil ECe chloride and SAR by depth	Shows root-zone accumulation and whether sodium may be impairing structure.
Emitter flow and pressure	Identifies dry salty zones and overwatered zones caused by poor uniformity.
Soil moisture and wetting depth	Shows whether irrigations refill the active roots and where the salt front is moving.
Drainage and restrictive layers	Determines whether leaching is feasible without prolonged saturation.
Root condition and rootstock	Separates exposure from uptake capacity disease and genetic tolerance.
Fertilizers and amendments	Finds hidden chloride sodium or EC additions.

Response options to evaluate

- **Reduce the source where feasible.** Test alternate water, blending, seasonal source changes, and all fertilizer or amendment inputs.
- **Improve uniformity.** Repair pressure or emitter problems so the driest trees are not receiving the largest salt concentration.
- **Manage timing and wetting pattern.** Maintain water availability to shallow roots while avoiding chronic saturation.
- **Leach only with a drainage plan.** Estimate need from water and soil tests, verify the water passes below roots, and monitor the response.
- **Avoid chloride-containing fertilizers when exposure is already high.** A nutrient source should not add the ion being managed.
- **Use rootstock and site history in long-term planning.** Tolerance cannot compensate for severe source water or drainage failure, but it can change risk.

Do not treat tip burn alone

The same grove can have chloride accumulation, sodium-related infiltration problems, poor drainage, and Phytophthora root rot at the same time. A complete diagnosis prevents one correction from worsening another problem.